

Discrete Structures Assignment 1

- 1.
- | | |
|-----------------------|-----------------------|
| a) Proposition: True | b) Proposition: False |
| c) Proposition: True | d) Proposition: False |
| e) Not a Proposition | f) Not a Proposition |
| g) Proposition: False | h) Proposition: True |
| i) Not a Proposition | j) Not a Proposition |

2. a) let $p = \text{"A student gets A in discrete"}$ and
 $q = \text{"his weightage is } \geq 95\% \text{"}$
 then $p \leftrightarrow q$.

b) let $p = \text{"Alice is smart"}$, $\neg p = \text{"she is not smart"}$,
 $q = \text{"honest"}$.
 then $p \vee (\neg p \wedge q)$.

c) let $p = \text{"}\sqrt{30} \text{ is greater than 6"}$, $q = \text{"}\sqrt{30} \text{ is less than 5"}$.
 then $p \oplus q$ (Exclusive OR) // I am assuming mathematical logic here

d) let $p = \text{"sam had pizza last night"}$, $q = \text{"chris finished her homework"}$.
 then $p \wedge q$

e) ~~let~~ let $p = \text{"chris finished her homework"}$, $q = \text{"pat watched news this morning"}$.
 then $p \oplus q$ (Exclusive OR)

3. a) two is greater than 5 and eight is an even integer. False

b) if 8 is an even integer and 11 is a prime number, then 2 is less than or equal to 5. True

c) if $2 \leq 5$, then 8 is an even integer or 11 is not a prime number.
 $\text{True} \rightarrow (\text{True} \vee (\neg \text{true})) \rightarrow \text{true} \rightarrow (\text{true} \vee \text{false}) \Rightarrow \text{true} \rightarrow \text{true} \Rightarrow \text{True}$

~~d) if 8 is not an even integer, then~~

d) if $2 \stackrel{>5}{\text{is not}}$, then 8 is not a even integer.

False $\rightarrow$ False $\Rightarrow$ True

4. a) (I will break the ^{statement} ~~comparison~~ into two comparisons joined by an AND)

let $P =$ "smartphone B has more RAM than smartphone A".

$q =$ "smartphone B has more RAM than smartphone C".

$\therefore P \wedge q, \Rightarrow \text{True} \wedge \text{True} \Rightarrow$ True.

b) let $P =$ "smartphone C has more ROM than smartphone B".

$q =$ "Smartphone C has higher resolution camera than smartphone B".

$\therefore P \vee q, \Rightarrow \text{false} \vee \text{True} \Rightarrow$ True.

c) Let $P =$ "smartphone B has more RAM than smartphone A".

$q =$ "smartphone B has more ROM than smartphone A".

$r =$ "smartphone B has higher resolution camera than smartphone A".

$\therefore P \wedge q \wedge r, \Rightarrow \text{True} \wedge \text{true} \wedge \text{false} \Rightarrow$ False.

d) let $P =$ "smartphone B has more ~~RAM~~ RAM than smartphone C".

$q =$ "smartphone B has more ROM than smartphone C".

$r =$ "It also has a higher resolution camera."

$\therefore (P \wedge q) \rightarrow r, (\text{True} \wedge \text{True}) \rightarrow \text{False} \Rightarrow \text{True} \rightarrow \text{False}$
 $\Rightarrow$ False.

e) let $P =$ "smartphone A has more RAM than smartphone B".

~~$q =$~~ "smartphone B has more RAM than smartphone A".

$\therefore P \leftrightarrow q, \text{False} \leftrightarrow \text{True} \Rightarrow$ False

5. a) let P = "Quixote Media had larger revenue than Acme computer".
 q = "Quixote Media had larger revenue than Nadir Software".
 $\therefore P \wedge q$, $\text{False} \wedge \text{True} \Rightarrow \underline{\text{False}}$

b) let P = "Nadir Software had lowest net profit".
 q = "Acme Computer had largest annual revenue".
 then $P \wedge q$, $\text{True} \wedge \text{True} \Rightarrow \underline{\text{True}}$

c) let P = "Acme Computer had the largest net profit".
 q = "Quixote Media had the largest net profit".
 then $P \oplus q$, (exclusive OR because they both cannot have
 $\text{False} \oplus \text{True}$ largest profit at once)
 $\Rightarrow \underline{\text{True}}$

d) let P = "Quixote Media had the smallest net profit".
 q = "Acme computer had the largest annual revenue".
 then: $P \rightarrow q$, $F \rightarrow T \Rightarrow \underline{\text{True}}$

e) let P = "Nadir Software had the smallest net profit".
 q = "Acme Computer had the largest annual revenue".
 then $P \leftrightarrow q$, $T \leftrightarrow T$, $\Rightarrow \underline{\text{True}}$.

6. a) "if you have the flue, then you miss the final examination".

b) "~~if~~ you don't miss the final examination ~~then~~ you pass the course."
 if and only if

c) "if you miss the final examination then you do not pass the course"

d) "~~if~~ you miss the final or

d) "You have the flue or you miss the final examination or you pass the course."

e) "if you have the flu then you ~~can~~ ^{don't} pass the course, or if you miss the final then you don't pass the course."

f) "~~if~~ you have the flu and miss the final, or you don't miss the final and pass the course."

7. a) $r \wedge \neg p$

b) $r \wedge q \wedge p$

c) ~~$r \rightarrow p$~~

d) $(p \wedge \neg q) \wedge r$

e) $(p \wedge q) \rightarrow r$

f) $r \leftrightarrow (p \vee q)$

8. a) if you send me an e-mail message, then I will remember to send you the address.

b) if you were born in the united states, then you are a citizen of this country.

c) if you keep your textbook, then it will be a useful reference in your future courses.

d) If their goalie plays well, The Red Wings will win the stanly cup.

e) If you get the job, then you had the best credentials.

f) if theres a storm, then the beach erodes.

g) if you log on the server, then you have a valid password.

h) If you don't begin your climb too late, then you will reach the summit.

9. a) 1. if it is sunny, ~~then~~ I will go for a walk in the woods.

2. It is sunny, that is sufficient for me to go for a walk in the woods.

3. I will go for a walk ^{in the woods} if it is sunny ^{tomorrow}.

4. I will go for a walk ^{in the woods} whenever it is sunny ^{tomorrow}.

5. I will go for a walk ^{in the woods} unless it is not sunny ^{tomorrow}.

6. Going on a walk ^{in the woods} is necessary for a sunny day ^{tomorrow}.

$q \rightarrow p$

b) Converse: If I go on a walk in the woods, then it is a sunny day.

Inverse: If it's not a sunny day tomorrow, then I will not

go ~~on~~ for a walk in the woods.

Contrapositive: I will not go for a walk in the woods if it is not sunny day tomorrow.

c) Inverse of it's Inverse: $\neg(\neg p \rightarrow \neg q) \Rightarrow p \rightarrow q$ (same as Conditional)

If it is sunny tomorrow, then I will go for a walk in the woods.

Inverse of it's Converse: $\neg(q \rightarrow p) \Rightarrow \neg q \rightarrow \neg p$ (contrapositive)

I will not go for a walk in the woods if it is not sunny tomorrow.

(converse)

Inverse of it's contrapositive: $\neg(\neg q \rightarrow \neg p) \Rightarrow q \rightarrow p$

If I go on a walk in the woods, then it is a sunny day.

10. De Morgan's Law: $\neg(p \wedge q) = \neg p \vee \neg q$, $\neg(p \vee q) = \neg p \wedge \neg q$ a) let p = "Jan is rich", q = "Jan is happy".we have $p \wedge q$, negation: $\neg(p \wedge q) \rightarrow \neg p \vee \neg q$
 $\neg p \vee \neg q$: "Jan is not rich or not happy."b) p = "carlos will bicyck", q = "Run tomorrow".we have $p \vee q \rightarrow \neg(p \vee q) \Rightarrow \neg p \wedge \neg q$
 $\neg p \wedge \neg q$: "Carlos will not bicyck and will not run tomorrow."c) p = "Fan is very slow", q = "it is very hot".we have $p \vee q \rightarrow \neg(p \vee q) \Rightarrow \neg p \wedge \neg q$ $\neg p \wedge \neg q$: "The fan is not slow and it is not very hot."

d) Here $P = \text{"Akram is unfit"} \wedge Q = \text{"Saleem is injured"}$.

$$P \wedge Q \rightarrow \neg(P \wedge Q) \rightarrow \neg P \vee \neg Q.$$

$\neg P \vee \neg Q$: "Akram is not unfit or Saleem is not injured."

e) Here $P = 0 \leq x \wedge x \leq 4$

$$P \wedge Q \rightarrow \neg(P \wedge Q) \rightarrow \neg P \vee \neg Q.$$

$$\neg P \vee \neg Q = (0 > x) \vee (x > 4)$$

f) Here $P = -2 \leq x \wedge Q = x \leq 5$

$$P \wedge Q \rightarrow \neg(P \wedge Q) \rightarrow \neg P \vee \neg Q$$

$$\neg P \vee \neg Q = (-2 > x) \vee (x > 5)$$

g) This is exact same as c.

h) Here $P = \text{"Akram is unfit"} \wedge Q = \text{"Ahmed is injured"}$.

$$P \wedge Q \Rightarrow \neg(P \wedge Q) \rightarrow \neg P \vee \neg Q.$$

$\neg P \vee \neg Q = \text{"Akram is not unfit and or Ahmed is not injured"}$.

11.

a) Exclusive OR b) Inclusive OR c) Inclusive OR d) Inclusive OR

12. a) $(P \wedge (\neg(\neg P \vee Q))) \vee (P \wedge Q) \equiv P$

~~$(P \wedge (\neg P \vee \neg Q)) \vee (P \wedge Q) \equiv P$~~

~~$(P \wedge P) \vee (P \wedge (\neg(\neg P \vee Q))) \vee (P \wedge Q) \equiv P$~~

Here $\neg(\neg P \vee Q)$, as De Morgan's Law: $\neg(A \vee B) = \neg A \wedge \neg B$

$\therefore \neg(\neg P \vee Q) \rightarrow P \wedge \neg Q$

and $\neg(A \wedge B) = \neg A \vee \neg B$

$\Rightarrow (P \wedge (P \wedge \neg Q)) \vee (P \wedge Q) \equiv P$

$\Rightarrow (P \wedge P \wedge \neg Q) \vee (P \wedge Q) \equiv P \rightarrow (P \wedge \neg Q) \vee (P \wedge Q)$

Here as $ab + ac = a(b + c)$, same Here

$P \wedge (\neg Q \vee Q)$

$(\neg Q \vee Q = T)$

$\Rightarrow P \wedge T \rightarrow P$ proven.

b) $\neg(P \leftrightarrow Q) \equiv (P \leftrightarrow \neg Q)$

$\neg(P \leftrightarrow Q)$ as $(P \leftrightarrow Q) \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$

$\neg((P \rightarrow Q) \wedge (Q \rightarrow P))$ as $(P \rightarrow Q) = \neg P \vee Q$

$\neg((\neg P \vee Q) \wedge (\neg Q \vee P))$ De Morgan's Law

$\neg(\neg P \vee Q) \vee \neg(\neg Q \vee P)$ De Morgan's Law

$(P \wedge \neg Q) \vee (Q \wedge \neg P)$ Distributive law of OR over AND

Here as $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$

$((P \wedge \neg Q) \vee Q) \wedge ((P \wedge \neg Q) \vee \neg P)$ Here $A = (P \wedge \neg Q)$, $B = Q$, $C = \neg P$

$(Q \vee P) \wedge (Q \vee \neg P) \wedge ((\neg P \vee P) \wedge (\neg P \vee \neg Q))$ Distributive law again

$((Q \vee P) \wedge T) \wedge (T \wedge (\neg P \vee \neg Q))$ Here as $P \vee \neg P = T$

$(Q \vee P) \wedge (\neg P \vee \neg Q)$ As $\neg(\neg P) = P$ & $\neg P \vee Q \equiv P \rightarrow Q$

$(\neg(\neg Q) \vee P) \wedge (\neg P \vee \neg(Q)) \rightarrow (\neg Q \vee P) \wedge (P \vee \neg Q)$

$\Rightarrow (P \rightarrow \neg Q) \wedge (\neg Q \rightarrow P)$

$\Rightarrow (P \rightarrow \neg Q) \wedge (\neg Q \rightarrow P)$ as $(P \leftrightarrow Q) = (P \rightarrow Q) \wedge (Q \rightarrow P)$

$\therefore P \leftrightarrow \neg Q$ proven.

c) $\neg P \leftrightarrow Q \equiv P \leftrightarrow \neg Q$

$\neg P \leftrightarrow Q$

as $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$

$(\neg P \rightarrow Q) \wedge (Q \rightarrow \neg P)$

as $P \rightarrow Q = (\neg P \vee Q)$

$(\neg(\neg P) \vee Q) \wedge (\neg Q \vee \neg P)$

$(P \vee Q) \wedge (\neg Q \vee \neg P)$

Commutative law $(\neg Q \vee \neg P) \rightarrow (\neg P \vee \neg Q)$

$(\neg Q \vee \neg P) \vee (P \vee Q)$

as $P \leftrightarrow Q \equiv (\neg P \vee Q)$

$(Q \vee \neg P) \vee (\neg(P \rightarrow Q))$

as $P \rightarrow Q = (P \rightarrow Q) \wedge (Q \rightarrow P)$

$(Q \vee P) \wedge (\neg P \vee \neg Q) \rightarrow (\neg P \vee \neg Q) \wedge (Q \vee P)$

as $P \rightarrow Q \equiv (\neg P \vee Q)$

$(P \rightarrow \neg Q) \wedge (\neg Q \rightarrow P)$

as $P \leftrightarrow Q = (P \rightarrow Q) \wedge (Q \rightarrow P)$

$(P \leftrightarrow \neg Q)$ proven.

$$d) (P \wedge Q) \rightarrow (P \rightarrow Q) \equiv T$$

$$(P \wedge Q) \rightarrow (\neg P \vee Q)$$

$$\text{as } P \rightarrow Q \equiv \neg P \vee Q$$

$$\rightarrow (P \wedge Q) \vee (\neg P \vee Q)$$

$$\text{as } P \rightarrow Q \equiv \neg P \vee Q$$

$$(\neg P \vee \neg Q) \vee (\neg P \vee Q)$$

$$\text{as } \neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg P \vee \neg Q \vee \neg P \vee Q$$

$$(\neg P \vee \neg P) \vee (\neg Q \vee Q)$$

$$\text{as } \neg P \vee P = T$$

$$\neg P \vee T \Rightarrow T \text{ proven.}$$

$$e) \neg(P \vee \neg(P \wedge Q)) \equiv F$$

$$\neg(P \vee (\neg P \vee \neg Q))$$

$$\text{as } \neg(P \wedge Q) = \neg P \vee \neg Q$$

$$\neg(P \vee \neg P \vee \neg Q)$$

$$\text{as } P \vee \neg P = T$$

$$\neg(T \vee \neg Q) \rightarrow \neg(T) \rightarrow F \text{ proven.}$$

$$13. a) (P \rightarrow R) \wedge (Q \rightarrow R) \text{ and } (P \vee Q) \rightarrow R$$

P	Q	R	$P \rightarrow R$	$Q \rightarrow R$	$P \vee Q$	$(P \rightarrow R) \wedge (Q \rightarrow R)$	$(P \vee Q) \rightarrow R$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	T	T	T	T	T
T	F	F	F	T	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F
F	F	T	T	T	F	T	T
F	F	F	T	T	F	T	T

They are logically equivalent.

b) $(P \rightarrow q) \vee (P \rightarrow r)$ and $P \rightarrow (q \vee r)$

P	q	r	$P \rightarrow q$	$P \rightarrow r$	$q \vee r$	$(P \rightarrow q) \vee (P \rightarrow r)$	$P \rightarrow (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	F	T	T

They are logically equivalent.

c) $(P \rightarrow q) \rightarrow (r \rightarrow s)$ and $(P \rightarrow r) \rightarrow (q \rightarrow s)$

P	q	r	s	$P \rightarrow q$	$r \rightarrow s$	$P \rightarrow r$	$q \rightarrow s$	$(P \rightarrow q) \rightarrow (r \rightarrow s)$	$(P \rightarrow r) \rightarrow (q \rightarrow s)$
T	T	T	T	T	T	T	T	T	T
T	T	T	F	T	F	T	F	F	F
T	T	F	T	T	T	F	T	T	T
T	T	F	F	T	T	F	F	T	T
T	F	T	T	F	T	T	T	T	T
T	F	T	F	F	F	T	T	T	T
T	F	F	T	F	T	F	T	T	T
T	F	F	F	F	T	F	T	T	T
F	T	T	T	T	T	T	T	T	T
F	T	T	F	T	F	T	F	F	F
F	T	F	T	T	T	T	T	T	T
F	T	F	F	T	T	T	F	<u>T</u>	<u>F</u>
F	F	T	T	T	T	T	T	T	T
F	F	T	F	T	F	T	T	F	T
F	F	F	T	T	T	T	T	<u>T</u>	<u>T</u>
F	F	F	F	T	T	T	T	T	T

They are NOT logically equivalent.

14. $q(x,y): x+y=1$

a) $q(x,y)$

$x+y=1$

Not a Proposition

b) $q(-6,7)$

$-6+7=1$

$1=1$

A True Proposition

c) $q(x+1, -x)$

$x+1-x=1$

$1=1$

A True Proposition

d) $q(x,3)$

$x+3=1$

Not a Proposition.

e) $q(1,1)$

$1+1=1$ A False Proposition

$2=1$

f) $q(5,-4)$

$5-4=1 \rightarrow 1=1$

A True Proposition.

A

15. $P(m,n): n = km$, $D \in \mathbb{Z}$

a) $P(4,5)$

$5 = k4 \rightarrow \frac{5}{4} = k$

False.

b) $P(2,4)$

$4 = k2 \rightarrow \frac{4}{2} = k$

$\rightarrow 2 = k$

True.

c) $\forall m \forall n P(m,n)$

False. Every positive integer does not divide every positive integer.

d) $\exists m \forall n P(m,n)$

True. One specific integer m (1) divides every integer n .

e) $\exists n \forall m P(m,n)$

False. one specific integer n that can be divided by every integer m (impossible).

f) $\forall n P(1,n)$

True. 1 divides every positive integer.

16. a) $\exists x (x^2=2)$

$x^2=2 \rightarrow x=\pm\sqrt{2}$

both are real numbers

and can be squared to squares to 2.

$\therefore$ True

b) $\exists x (x^2=-1)$

False.

No real number

squares to -1.

c) $\forall x (x^2+2 \geq 1)$

True. any real number, negative or positive, will satisfy the equation.

d) $\exists x (x^2=x)$

True. Both 0 and 1

satisfy the equation.

17. $F(x, y) = "x \text{ can fool } y"$.

a) $\forall x F(x, \text{Bob})$.

b) $\forall y F(\text{Alice}, y)$.

c) $\forall x \exists y F(x, y)$.

d) $\neg \exists x \forall y F(x, y)$.

e) $\exists x \forall y F(x, y)$.

18. $P(x) = "x \text{ can speak russian}"$, $Q(x) = "x \text{ knows the computer language } C++"$.

a) $\exists x (P(x) \wedge Q(x))$.

b) $\exists x (P(x) \wedge \neg Q(x))$.

c) $\forall x (P(x) \vee Q(x))$.

d) $\neg \exists x (P(x) \wedge Q(x))$.

19. $Q(x, y) = "x \text{ has sent an e-mail message to } y"$.

a) $\exists x \exists y Q(x, y)$

There exists a student who has sent an email message to someone.

b) $\exists x \forall y Q(x, y)$

There is a student who has sent an email message to everyone.

c) $\forall x \exists y Q(x, y)$

Every student has sent an e-mail message to someone.

d) $\exists x \forall y Q(x, y)$

There is a student who ^{was} ~~has~~ sent e-mail messages by everyone.

e) $\forall y \exists x Q(x, y)$

Every student was sent e-mail messages by someone.

f) $\forall x \forall y Q(x, y)$

Every student has sent email messages to everyone.

20. $P(x, y) = "student x \text{ has taken class } y"$

Domain of x : All ~~set~~ students in the class.

Domain of y : All computer science courses in school.

a) $\exists x \exists y P(x, y)$

Some students has taken some computer science course classes.

b) $\exists x \forall y P(x, y)$

Some students have taken all computer science course classes.

c) $\forall x \exists y P(x, y)$

At Every student has taken some computer science course classes.

d) $\exists y \forall x P(x, y)$

Some computer science course classes ^{were} ~~where~~ taken by all students.

e) $\forall y \exists x P(x, y)$

All computer science course classes were taken by some students.

f) $\forall x \forall y P(x, y)$

All students have taken all computer science classes.

21.

a) Addition b) Simplification c) Modus Ponens

d) Modus Tollens e) Hypothetical Syllogism

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a) let p = today is tuesday, q = test in mathematics, r = test in economics, s = my economics professor is sick

First statement: $P \rightarrow (P \vee q \vee r)$ Third: $P \wedge S$

2nd statement: $S \rightarrow \neg r$

~~$P \rightarrow (P \vee q \vee r)$~~ using Addition ~~$(P \vee q \vee r) \Rightarrow P \rightarrow (q \vee r)$~~
 ~~$P \rightarrow q$~~ using Modus Ponens ~~$(P \rightarrow q, P \Rightarrow q)$~~

since third is true, we can use Simplification ($P \wedge Q \therefore P$)

$$\therefore P \wedge S \Rightarrow S \text{ and } P.$$

Now first ~~stat~~ premise: $P \rightarrow (q \vee r)$, and we have P

using Modus Ponens $\Rightarrow q \vee r$.

second premise $S \rightarrow \neg r$, we have S .

using Modus Ponens $\Rightarrow \neg r$

Now we have $q \vee r$, $\neg r$

using ~~Modus Tollens~~ ($\neg P, P \vee Q \therefore Q$) Disjunctive syllogism ($P \vee Q, \neg P \therefore Q$)

$\Rightarrow q$ = test in mathematics.

b) let P = "Ali is a lawyer", q = "Ali is ambitious",

r = "Ali is an early riser", s = "Ali likes chocolates"

1st Premise: $P \rightarrow q$

2nd Premise: $r \rightarrow \neg s$

3rd Premise: $q \rightarrow r$

we have 1st and 2nd Premise: $P \rightarrow q, q \rightarrow r$

using Hypothetical syllogism ($P \rightarrow q, q \rightarrow r \Rightarrow P \rightarrow r$) $\Rightarrow P \rightarrow r$.

* Now we have $P \rightarrow r$ and $r \rightarrow \neg s$

using Hypothetical syllogism again $\Rightarrow P \rightarrow \neg s$

& $P \rightarrow \neg s$: if Ali is a Lawyer, then he does not like chocolates.

c) $((t \rightarrow (r \vee p)) \rightarrow ((\neg r \vee k) \wedge \neg k)) \rightarrow \neg r$

Premise: $((t \rightarrow (r \vee p)) \rightarrow ((\neg r \vee k) \wedge \neg k))$

Conclusion: $\neg r$

So to prove the above, we will use Proof by Contradiction method. Because the above single conditional does not allow us to easily extract variables.

We will temporarily assume the opposite of our goal ($\neg r$) is true.

$r = T$. (assumption), creating the left premise using this.

~~$t \rightarrow (r \vee p)$~~ $r \Rightarrow$ Addition ($P \therefore P \vee Q \Rightarrow r \vee p \Rightarrow \neg t \vee (r \vee p)$)

as $P \rightarrow Q \equiv \neg P \vee Q \Rightarrow t \rightarrow (r \vee p)$, left side created.

Now we can use it in Modus Ponens.

we have $t \rightarrow (r \vee p)$, $(t \rightarrow (r \vee p)) \rightarrow ((\neg r \vee k) \wedge \neg k)$

using Modus Ponens, we get

$$((\neg r \vee k) \wedge \neg k)$$

using Simplification rule to detach both sides

$$\neg k \quad \text{and} \quad \neg r \vee k$$

Now we can use Disjunctive Syllogism ($p \vee q, \neg p \therefore q$)

$$\Rightarrow \neg r$$

(Assumption)

Now we have r and $\neg r$. One of these must be

true. using conjunction ($p, q \therefore p \wedge q$)

$$r \wedge \neg r$$

The above is mathematically impossible, and all the laws used were correct. So the only error must be our assumption that r is true. $\therefore r$ is false and $\neg r$ must be true.

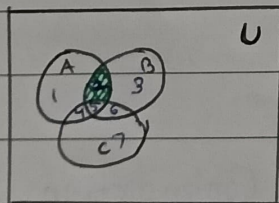
$$23. U = \{1, 2, 3, 4, 5, 6, 7, 8\}, A = \{1, 2, 4, 5\}, B = \{2, 3, 5, 6\}, C = \{4, 5, 6, 7\}$$

$$a) (A \cap B) \cap \bar{C}$$

$$A \cap B = \{2, 5\}, \bar{C} = \{1, 3, 8\}$$

$$\bar{C} = \{1, 2, 3, 8\}$$

$$(A \cap B) \cap \bar{C} = \{2\}$$

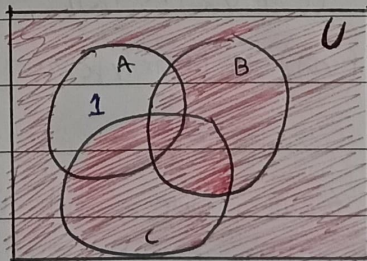


$$b) \bar{A} \cup (B \cup C)$$

$$\bar{A} = \{3, 6, 7, 8\}$$

$$B \cup C = \{2, 3, 4, 5, 6, 7\}$$

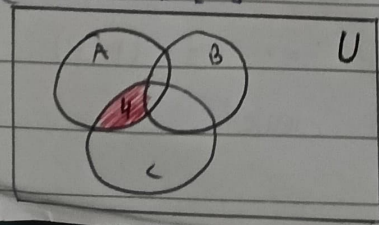
$$\bar{A} \cup (B \cup C) = \{2, 3, 4, 5, 6, 7, 8\}$$



$$c) (A - B) \cap C$$

$$A - B = \{1, 4\}$$

$$(A - B) \cap C = \{4\}$$



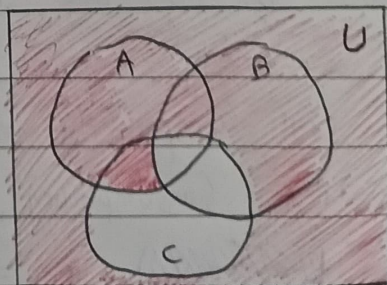
$$d) (A \cap \bar{B}) \cup \bar{C}$$

$$\bar{B} = \{1, 4, 7, 8\}$$

$$(A \cap \bar{B}) = \{1, 4\}$$

$$\bar{C} = \{1, 2, 3, 8\}$$

$$(A \cap \bar{B}) \cup \bar{C} = \{1, 2, 3, 4, 8\}$$



24. a) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$

A is subset of B: $A \subset B$

A is a proper subset of B: $A \subsetneq B$

b) $C = \{x/x \text{ is a prime number less than } 10\}$

All prime numbers less than 10: $C = \{2, 3, 5, 7\}$

Proper subsets: $\emptyset, \{2\}, \{3\}, \{5\}, \{7\}, \{2, 3\}, \{2, 5\}, \{2, 7\}, \{3, 5\}, \{3, 7\}, \{5, 7\}, \{2, 3, 5\}, \{2, 3, 7\}, \{2, 5, 7\}, \{3, 5, 7\}$.

c) $L = \{1, 2, 3\}$

All subsets: $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

$P(L) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

d) $M = \text{set of } k \text{ elements.}$

Number of elements in the power set of a set with k element is 2^k .

e) $N = \{x/x \text{ is a prime number between } 10 \text{ and } 20\}$.

$N = \{11, 13, 17, 19\}$. The cardinality of N (num of elements) is 4.

$\therefore$ Cardinality of power set is $2^4 = 16$.

f) $E = \{1, 2, 4\}$, $F = \{a, b, c, d\}$

Cartesian product pairs every element of first set with every element of the second set.

$E \times F = \{(1, a), (1, b), (1, c), (1, d), (2, a), (2, b), (2, c), (2, d), (4, a), (4, b), (4, c), (4, d)\}$

25. a) $(A - (A \cap B)) \cap (B - (A \cap B)) = \emptyset$

$(A \cap (\overline{A \cap B})) \cap (B \cap (\overline{A \cap B}))$

$\therefore A - B = A \cap \overline{B}$

$(A \cap (\overline{A} \cup \overline{B})) \cap (B \cap (\overline{A} \cup \overline{B}))$

$\Rightarrow A \cap (\overline{A} \cup \overline{B}) \cap B \cap (\overline{A} \cup \overline{B}) \Rightarrow (A \cap B) \cap (\overline{A} \cup \overline{B}) \cap (\overline{A} \cup \overline{B}) \therefore A \cap A = A$

$\Rightarrow (A \cap B) \cap (\overline{A} \cup \overline{B}) \quad \text{as } A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$$((A \cap B) \cap \bar{A}) \cup ((A \cap B) \cap \bar{B}) \rightarrow (A \cap B \cap \bar{A}) \cup (A \cap B \cap \bar{B}) \quad \therefore A \cap \bar{A} = \phi$$

$$(\phi \cap B) \cup (A \cap \phi) \rightarrow \phi \cup \phi \rightarrow \phi.$$

$$b) (A-B) \cup (A \cap B) = A$$

$$(A \cap \bar{B}) \cup (A \cap B) \neq A$$

$$A \cap (\bar{B} \cup B)$$

$$A \cap U = A.$$

$$\therefore A-B = A \cap \bar{B}$$

$$\therefore A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$\therefore \bar{B} \cup B = U, \quad A \cap U = A.$$

$$c) (A-B)-C = (A-C)-B$$

$$(A \cap \bar{B}) \cap \bar{C}$$

$$(A \cap \bar{C}) \cap \bar{B}$$

$$(A-C)-B.$$

$$\therefore A-B = A \cap \bar{B}$$

$$\therefore (A \cap \bar{B}) \cap C = A \cap (B \cap C)$$

$$\therefore A-B = A \cap \bar{B}$$

$$d) (B \cup (B-A)) = B \quad (\text{Question is glitched in the file})$$

$$B \cup (B \cap \bar{A})$$

$$\Rightarrow B.$$

$$\therefore A-B = A \cap \bar{B}$$

$$\text{As } A \cap (A \cap B) = A \quad (\text{absorption law})$$

$$26. a) U = 100, \text{ Worms } |W| = 20, \text{ Bruised } |B| = 15, |W \cap B| = 10.$$

$$\text{As } |A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B| = 20 + 15 - 10 \rightarrow 25.$$

(total apples
that can't be sold)

$$\text{Total} - A \cup B \Rightarrow 100 - 25 \Rightarrow 75 \text{ apples that can be sold.}$$

$$b) U = 1000, |C| = 350, |S| = 450, |C \cap S| = 100$$

To find students that like either, we need $|A \cup B|$

$$|A \cup B| = |A| + |B| - |A \cap B| \rightarrow 350 + 450 - 100 \rightarrow 700.$$

To find students that don't like either: Total - liked

$$1000 - 700 \Rightarrow 300 \text{ students.}$$

c) Mixed Berry $|M| = 78$, Irish Cream $|I| = 32$, Tiramisu $|T| = 57$.

$$|M \cap I| = 13, |I \cap T| = 21, |T \cap M| = 16, |M \cap I \cap T| = 5.$$

$$\text{As } A \cup B \cup C = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$A \cup B \cup C = 78 + 32 + 57 - 13 - 21 - 16 + 5 = 122.$$

14 students didn't like any.

$$\therefore U = 122 + 14 \rightarrow U = 136 \text{ total students}$$

$$d) A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Translating left side into set builder.

$$A \times (B \cap C) = \{(x, y) \mid x \in A \wedge y \in (B \cap C)\}$$

$$= \{(x, y) \mid x \in A \wedge (y \in B \wedge y \in C)\} \quad \because \cap \rightarrow \wedge$$

$$= \{(x, y) \mid (x \in A \wedge (y \in B \wedge y \in C)) \wedge (x \in A \wedge (y \in B \wedge y \in C))\} \quad \because P \equiv P \wedge P$$

$$= \{(x, y) \mid (x, y) \in (A \times B) \wedge (x, y) \in (A \times C)\} \quad \text{converting back into}$$

now translating $\wedge \rightarrow \cap$

cartesian notation

$$\Rightarrow (A \times B) \cap (A \times C).$$

$$27. A = \{a, a, a, b, b, c\}$$

$$m_A(a) = 3, m_A(b) = 2, m_A(c) = 1, m_A(d) = 0, \text{ Total} = 6.$$

28.

$$\{a, a, b, c\} \text{ and } \{a, b, a, c\} : \text{Equal}$$

$$\{1, 1, 2, 3\} \text{ and } \{1, 2, 2, 3\} : \text{Not Equal}$$

$$\{x, x, x, y\} \text{ and } \{x, y, x, x\} : \text{Equal}$$

$$29. A = \{a, a, b, c\}, B = \{a, b, b, c, c\}$$

(This is pretty simple so

$$A \cup B = \{a, a, b, b, c, c\}$$

I won't write each step)

$$A \cap B = \{a, b, c\}$$

$$A \oplus B = \{a, a, a, b, b, b, c, c, c\}$$

$$A - B = \{a\}$$

$$B - A = \{b, c\}$$

30. let r = red balls, b = blue balls, g = green balls
 $A = \{r, r, r, b, b, g, g, g, g\}$

31. let n = Notebooks, p = pens, c = pencil
 $L = \{n, n, n, p, p, c\}$

32. $A = \{\text{apple, apple, apple, orange, orange}\}$
 $B = \{\text{apple, apple, banana, orange}\}$

$A \cup B = \{\text{apple, apple, apple, orange, orange, banana}\}$

$A \cap B = \{\text{apple, apple, orange}\}$.